

6.4 Gram–Schmidt and QR Decomposition

Objectives: Topics

- The Gram–Schmidt Process
- The QR decomposition

Objectives: Goals

- Apply Gram–Schmidt and QR decomposition to construct orthogonal or orthonormal bases.
- Compute the QR factorization of a matrix.

Motivating Question

Identify an orthogonal basis for the subspace spanned by:

$$\vec{x}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad \vec{x}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad \vec{x}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}.$$

Theorem 6.44. The Gram–Schmidt Process. Given a basis $\{\vec{x}_1, \dots, \vec{x}_p\}$ for a subspace $W \subseteq \mathbb{R}^n$, define:

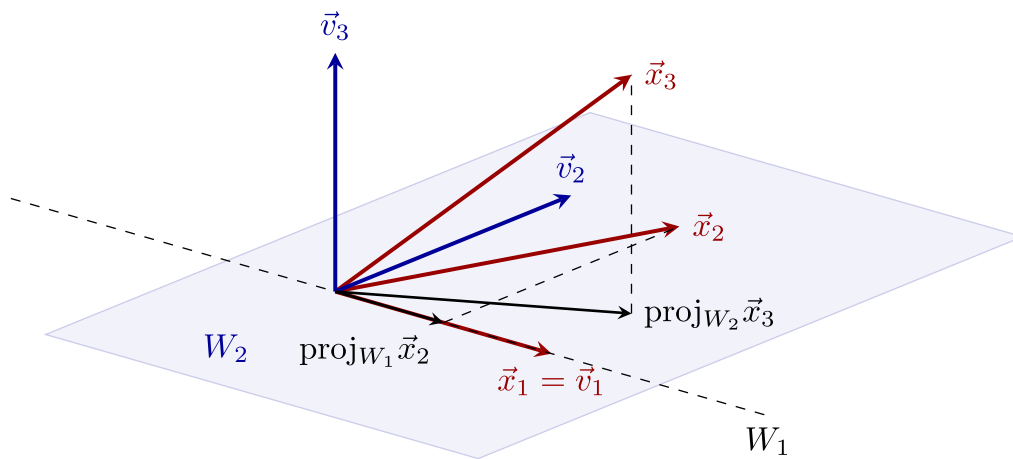
$$\begin{aligned} \vec{v}_1 &= \vec{x}_1, \\ \vec{v}_2 &= \vec{x}_2 - \frac{\vec{x}_2 \cdot \vec{v}_1}{\vec{v}_1 \cdot \vec{v}_1} \vec{v}_1, \\ \vec{v}_3 &= \vec{x}_3 - \frac{\vec{x}_3 \cdot \vec{v}_1}{\vec{v}_1 \cdot \vec{v}_1} \vec{v}_1 - \frac{\vec{x}_3 \cdot \vec{v}_2}{\vec{v}_2 \cdot \vec{v}_2} \vec{v}_2, \\ &\vdots \\ \vec{v}_p &= \vec{x}_p - \frac{\vec{x}_p \cdot \vec{v}_1}{\vec{v}_1 \cdot \vec{v}_1} \vec{v}_1 - \dots - \frac{\vec{x}_p \cdot \vec{v}_{p-1}}{\vec{v}_{p-1} \cdot \vec{v}_{p-1}} \vec{v}_{p-1}. \end{aligned}$$

Then $\{\vec{v}_1, \dots, \vec{v}_p\}$ is an orthogonal basis for W . Additionally,

$$W_k = \text{Span}\{\vec{v}_1, \dots, \vec{v}_k\} = \text{Span}\{\vec{x}_1, \dots, \vec{x}_k\} \quad \text{for } 1 \leq k \leq p.$$

Thus, we see that each vector $\vec{v}_k$ is formed by starting with $\vec{x}_k$ and then removing the orthogonal projection of $\vec{x}_k$ onto W_{k-1} ; that is,

$$\vec{v}_k = \vec{x}_k - \text{proj}_{W_{k-1}} \vec{x}_k = \vec{x}_k - \text{proj}_{\vec{v}_1} \vec{x}_k - \dots - \text{proj}_{\vec{v}_{k-1}} \vec{x}_k.$$



①

Vectors $\vec{x}_1, \vec{x}_2, \vec{x}_3$ are used to produce orthogonal vectors $\vec{v}_1, \vec{v}_2, \vec{v}_3$ via subtraction of projections.

Gram-Schmidt successively removes the components of each vector in the span of earlier ones.

Example 6.45. Let $W = \text{Span} \left\{ \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 8 \\ 5 \\ -6 \end{bmatrix} \right\}$. Find an orthogonal basis for W .